Complex Analysis Homework 6

1. (a) Suppose that $f(z)$ is analytic on and inside the closed disk $|z| \le R$ and that $f(0) = 0$.

(i) Prove that the function

$$F(z) := \begin{cases} f(z)/z & z \neq 0 \\ f'(0) & z = 0 \end{cases}$$

is analytic on $|z| \le R$.

(ii) Suppose $|f(z)| \le M = \max_{|\omega|=R}\{|f(\omega)|\}$ inside the circle $|z| = R$. Show that

$$|f(z)| \le \frac{M}{R}|z| \quad \text{and} \quad |f'(0)| \le \frac{M}{R},$$

and equality holds only if $f(z) = \frac{M}{R}e^{i\theta}z,$ where θ is real.

(b) Suppose $f(z)$ is analytic on and inside the unit circle $|z| = 1$, and that $\text{Re}\, f(z) > 0,$ $f(0) = a > 0$, show that $\left|\frac{f(z)-a}{f(z)+a}\right| \le |z|$ and $|f'(0)| \le 2a.$

Hint: Consider the function $g(z) = \frac{z-a}{z+a}.$ Show that $|g(z)| < 1$ if $\text{Re}(z) > 0.$

2. Suppose f is analytic in $|z| \le 1$. Suppose $|f(z)| \le 2$ for $z = e^{i\theta},$ $0 < \theta \le \pi$ and $|f(z)| \le 3$ for $z = e^{i\theta},$ $-\pi < \theta \le 0.$ Show that $|f(0)| \le \sqrt{6}.$ [Hint: Consider $f(z) \cdot f(-z).$]

3. (a) Let α be a complex number. Show that if $(1+z)^{\alpha}$ is taken as $e^{\alpha \text{Log}(1+z)},$ then for $|z| < 1$

$$(1+z)^{\alpha} = 1 + \frac{\alpha}{1}z + \frac{\alpha(\alpha-1)}{2\cdot 1}z^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3\cdot 2\cdot 1}z^3 + \cdots.$$

(b) Give the Taylar series expansion of $\sqrt{z}$ at $z_0 = 1$ and find its radius of convergence.

4. Find the radius of convergence of the following power series:

(a) $\sum_{k=0}^{\infty} 2^k z^k$

(b) $\sum_{k=0}^{\infty} \frac{k}{6} z^k$

(c) $\sum_{k=1}^{\infty} k^2 z^k$

(d) $\sum_{k=0}^{\infty} \frac{3^k z^k}{4^k + 5^k}$

(e) $\sum_{k=1}^{\infty} \frac{2^k z^{2k}}{k^2 + k}$

(f) $\sum_{k=1}^{\infty} \frac{z^{2k}}{4^k k^k}$

(g) $\sum_{k=1}^{\infty} \frac{k^k}{1 + 2^k k^k} z^k$

5. Find the maximum and minimum absolute value of $z^2 - z$ in the disc: $|z| \leq 1$.